# Cleaning

### 01.Burets

Thoroughly clean the tube of the buret with detergent and a long brush. After being cleaned, buret should be thoroughly clean with tap water and then with three or four portions of distilled water. If necessary brief soaking in a warm detergent solution could be used to remove grease & dirt responsible for water breaks. (Note: Prolonged soaking should be avoided because a rough area or ring is likely to develop, at a detergent/air interface. Inspect for water breaches & if necessary repeat the above cleaning process.)

### 02.Pipet

Use a rubber bulb to detergent solution to a level 2–3 cm above the calibration mark of the pipet. Drain the solution & then rinse the pipet. Drain this solution & then rinse the pipet with several portions of tap water. Inspect for film breahs; if necessary, repeat the above cleaning process. Finally, fill the pipet with distilled water to perhaps 1/3 of its capacity & carefully rotate it so that entire interior surface is wetted. Repeat this rinsing step at least twice. To dry the pipet place it in an inverted position in the funnel holder of a funnel stand.

### 03. Volumetric flask

Wash with detergent & then thoroughly wash it with distilled water. To dry the flask, clamp it in an inverted position.

**Measurement of an Aliquot using pipette**

- Use a pipette filler to draw a small volume of the liquid to be sampled into the pipette.
- Thoroughly wet the entire interior surface.
- Repeat with at least two additional portion.
- Carefully fill the pipette to a level above the graduation mark.(Figure 1-A)
- Quickly replace the bulb with a forefinger to arrest the outflow of liquid.(Figure 1-B).Make sure that are no bubbles in the bulk of the liquid or foam at the surface.
- Till the pipette slightly from the vertical & wipe the exterior free of adhering liquid. (Figure 1-C)
- Touch the top of the pipette to the wall of a glass container (not the container into which the aliquot is to be transferred) & slowly allow the liquid level to drop by partially releasing the forefinger (Figure 1 –D). Halt further flow as the bottom of the meniscus coincides exactly with the graduation mark.
- Place the pipette tip well within the receiving flask & allow the liquid to drain.
- When free flow ceases, rest the tip of the pipette, against the inner wall of the receiver for 10 seconds (Figure 1-E)
- Withdraw the pipette with a rotating motion to remove any liquid adhering to the tip.

**Note:** Small volume retaining inside the tip of the pipette should not be blown or rinsed into the receiving flask.

Quantitative transfer of liquid to a volumetric flask

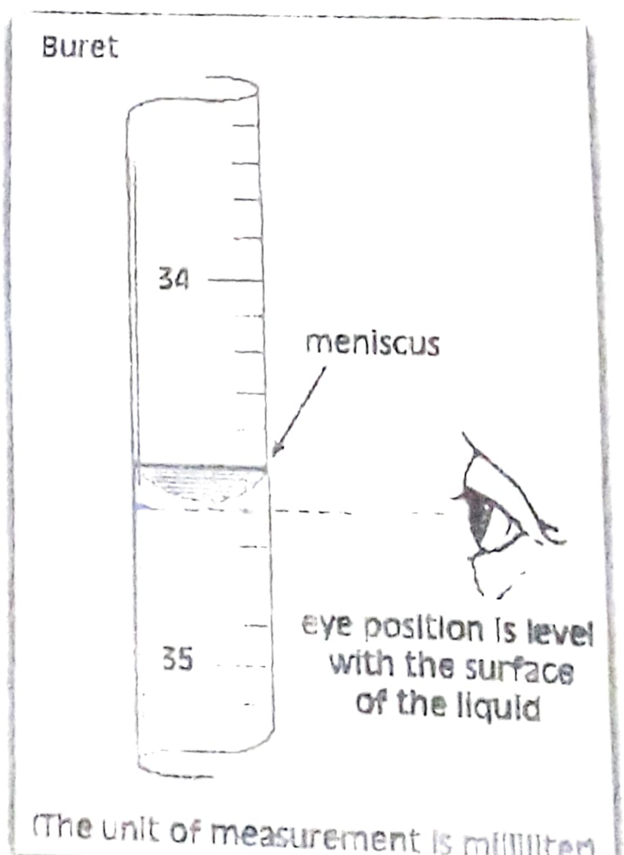
- Insert a funnel into the neck of the volumetric flask.
- Use a stirring rod to direct the flow of liquid from the beak flask into the funnel.
- Tip off the last drop of the liquid on the spout of the beaker with the stirring rod.
- Rinse both the stirring rod & the interior of the beaker with distilled water & transfer the washings to the volumetric flask, as before.
- Repeat the rinsing process at least two more times.

## Directions for the use of a burette

i. **Test for leakage:** Fill the buret with water and check whether volume reading changes with time.

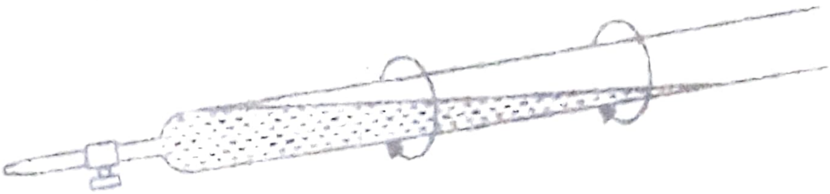
ii. **Drying:** Open the stopcock & clamp the buret in an inverted position

iii. **Method for reading a buret:** The eye should be level with the meniscus



iv. **Filling:**

- Close the stopcock.
- Add 5-10 $cm^3$ of the titrant & carefully rotate the buret to wet the interior completely.



Rinsing the burette

- Allow the liquid to drain through the tip.

Repeat this step at least two more times.

- Then fill the buret well above the zero mark.
- Remove air bubbles, by rapidly rotating the stopcock & permitting small quantities of the titrant to pass.
- Lower the level of the liquid to the zero mark.
- Allow for drainage (~1 min) & then record the initial volume reading (estimating to the nearest 0.05 $cm^3$)
- Figure 2 indicates the best method to handle the stopcock.

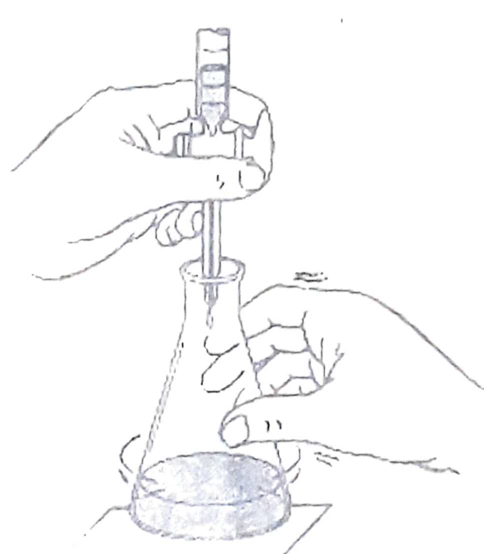


**Figure 2 Positioning the buret for titration and manipulating the stopcock.**